A Comparative Study of Rule-Based, Machine Learning, and Hybrid Dynamic Pricing Models for Stable and Customer-Centric Pricing in Retail SMEs

Introduction

Dynamic pricing has become increasingly prevalent in modern retail as machine-learning systems enable prices to adjust in response to demand patterns, competitor activity and contextual factors (Kopalle *et al.*, 2023). These technologies reduce manual intervention and allow retailers to respond more rapidly to market conditions (Jonwal, 2025). However, SMEs remain cautious due to limited data availability and concerns about customer trust, as frequent or abrupt price changes can disrupt reference prices and influence purchasing behaviour (Nouri-Harzvili and Hosseini-Motlagh, 2023). This study proposes a hybrid dynamic pricing approach that retains predictive value while maintaining pricing stability in SME contexts.

Background to the Study

Dynamic pricing models have largely been developed for data-rich sectors such as airlines, hospitality, and large e-commerce platforms (Kopalle *et al.*, 2023). These environments rely on continuous behavioural and market data, supported by optimisation infrastructures that smaller retailers typically lack. SME datasets are often irregular or low-volume (Kumari *et al.*, 2023), reducing the reliability of conventional pricing models. Prior research highlights behavioural sensitivity to price volatility and the interaction between pricing and operational constraints, revealing a gap in dynamic pricing research that motivates the need for stability-focused approaches tailored to SMEs (Nouri-Harzvili and Hosseini-Motlagh, 2023; Chen, Yang and Xu, 2021; Wang, Liu and Yang, 2023).

Problem Statements

Dynamic pricing research has predominantly focused on data-rich environments such as large e-commerce and airline platforms, where continuous behavioural and transactional data support complex optimisation and machine-learning models (Kopalle *et al.*, 2023). In contrast, SMEs often operate with limited or fragmented datasets, which undermines the stability and reliability of machine-learning-based pricing and increases the risk of unpredictable outcomes (Kumari *et al.*, 2023). Retail SMEs also face resource constraints, including limited analytical capacity, restricted budgets and low digital maturity, making it difficult to adopt and sustain sophisticated pricing systems (Gouveia and Mamede, 2022). Frequent or abrupt algorithmic price changes can disrupt customer reference prices and reduce trust by negatively affecting perceived fairness and purchasing behaviour (Nouri-Harzvili and Hosseini-Motlagh, 2023). These challenges are compounded by demand uncertainty, inventory interactions and capital constraints in SME retail settings (Wang, Liu and Yang, 2023).

Research Aim

This research aims to design and comparatively evaluate rule-based, machine-learning, and hybrid dynamic pricing models to assess their impact on price stability, customer centricity, and revenue performance in retail SMEs.

Research Objectives

1. To analyse dynamic pricing in retail SMEs in order to identify data constraints and behavioural factors that influence pricing stability and customer acceptance.
2. To implement rule-based, machine-learning, and hybrid dynamic pricing models using publicly available retail datasets.
3. To evaluate and compare the three pricing models in terms of price stability and revenue performance.
4. To examine how stability-oriented constraints influence pricing behaviour within the hybrid model relative to the other approaches.

Research Question

How does a hybrid dynamic pricing model compare with rule-based and machine-learning approaches in terms of price stability and revenue performance for retail SMEs?

Research Hypotheses

This study adopts a hypothesis-driven approach to support the comparative evaluation of dynamic pricing models in retail SME contexts.

Null Hypothesis (H_0):

There is no meaningful difference in price stability or revenue performance between rule-based, machine-learning, and hybrid dynamic pricing models when applied to retail SME datasets.

Alternative Hypothesis (H_1):

The hybrid dynamic pricing model produces more stable pricing outcomes than rule-based and machine-learning-only models, while maintaining comparable revenue performance in retail SME settings.

Significance of the Study

Dynamic pricing can improve revenue, inventory management and demand responsiveness (Wang, Liu and Yang, 2023), but the data and monitoring requirements of most systems limit adoption by SMEs due to constrained datasets and analytical capacity (Kumari *et al.*, 2023). Volatile pricing can further weaken customer acceptance by increasing perceived financial risk (Chen, Yang and Xu, 2021). A hybrid model that combines predictive pricing with stabilising constraints offers a practical alternative, supporting responsiveness while maintaining predictable pricing behaviour and customer trust (Kopalle *et al.*, 2023).

Rationale of the Study

This study is motivated by the misalignment between dominant dynamic pricing research and SME operational realities. While algorithmic pricing has demonstrated benefits in data-rich contexts, existing models are often unsuitable for SMEs due to structural and practical constraints, limiting their real-world applicability.

The study is additionally motivated by an academic interest in applied machine learning, as the use of predictive pricing methods in smaller retail settings remains underexplored. Evaluating a hybrid approach offers practical and academic value by demonstrating how predictive models can be adapted for low-data, resource-constrained environments.

Key Literature

The literature selected for this study draws on peer-reviewed academic and practitioner sources to establish theoretical grounding for dynamic pricing in retail SME environments. Academic sources focus on pricing models, machine-learning approaches, behavioural responses and operational constraints, while practitioner sources provide contextual implementation insight. Together, this literature informs the research problem, supports the formulation of the research question and hypotheses, and justifies the comparative evaluation undertaken in this study.

Chen, L., Yang, Y. and Xu, Q. (2021)

This study shows that excessive price volatility increases perceived financial risk and reduces acceptance of dynamic pricing, highlighting the importance of stability in pricing model design.

Data Science PM (2024)

This practitioner source outlines Agile Data Science principles and supports the use of iterative, flexible workflows for computational and modelling-based research projects.

Devarajanayaka, K.M. *et al.* (2024)

This paper demonstrates that machine-learning-based pricing optimisation can improve performance in online retail when sufficient historical data are available, while highlighting limitations in low-data environments.

Gubkin, A. (2023)

This industry article provides an overview of rule-based and algorithmic dynamic pricing models, supporting background understanding of pricing approaches used in retail systems.

Gouveia, F.D. and Mamede, H.S. (2022)

This study identifies organisational, analytical and resource constraints that limit digital transformation in retail SMEs, reinforcing the need for interpretable and resource-appropriate pricing systems.

Jonwal, S. (2025)

This practitioner article discusses the use of AI-driven pricing algorithms in modern retail and provides contextual insight into motivations and adoption considerations for automated pricing.

Kopalle, P.K. *et al.* (2023)

This comprehensive review highlights strategic, behavioural and technological aspects of dynamic pricing research and notes that much of the literature assumes data-rich environments, supporting the need for SME-adapted models.

Kumari, A. *et al.* (2023)

This study identifies limited data availability, system complexity and integration challenges as key barriers to implementing dynamic pricing systems in smaller retail firms.

Nouri-Harzvili, M. and Hosseini-Motlagh, S-M. (2023)

This research shows that frequent and abrupt price changes negatively affect perceived fairness and purchasing behaviour, motivating the inclusion of stability mechanisms in pricing models.

Yang, H. *et al.* (2021)

This study examines pricing decisions under demand uncertainty and capital constraints, highlighting how operational limitations influence pricing outcomes in resource-constrained retail settings.

Yoshi, A.M. *et al.* (2025)

This research demonstrates the potential of advanced machine-learning techniques for real-time dynamic pricing in e-commerce, while reinforcing assumptions about data availability that may limit applicability to SMEs.

Wang, X., Liu, S. and Yang, T. (2023)

This study shows that pricing strategies are closely linked to inventory dynamics and demand behaviour, supporting the use of revenue-based metrics in pricing evaluation.

Methodology

This study adopts an Agile Software Development Life Cycle using a Kanban workflow, which supports iterative refinement in computational research where modelling decisions evolve as insights are gained from data exploration (Data Science PM, 2024). This approach is appropriate for a single-researcher project in which analytical and modelling tasks must remain flexible rather than follow a fixed sequential structure.

The study will begin with the preparation of publicly available retail datasets through cleaning and feature engineering. Evaluation will be conducted by comparing the outputs of rule-based, machine-learning-only, and hybrid pricing approaches under consistent experimental conditions. Revenue-related indicators will be used to assess pricing performance, while volatility measures will be applied to evaluate price stability. Forecasting accuracy metrics will be used to assess the machine-learning component, and visual outputs will support the interpretation of differences in pricing behaviour across the models.

Artefact

The artefact produced by this research will be a standalone dynamic pricing engine developed as a research prototype to support the comparative evaluation of pricing approaches in retail SME contexts. The artefact will be implemented in Python and will operate exclusively on publicly available retail datasets, without interaction with live commercial systems or real-world pricing environments.

The pricing engine will implement three distinct models. A rule-based pricing model will provide a stable and interpretable baseline reflecting common SME pricing practices. A machine-learning-based pricing model will generate demand forecasts from historical data and produce dynamic price recommendations without explicit stabilisation mechanisms. A hybrid pricing model will integrate stability-oriented constraints, such as bounded price changes and smoothing mechanisms, with machine-learning forecasts in order to reduce unnecessary price volatility.

The artefact will generate structured outputs that enable direct comparison of pricing behaviour across the three approaches. These outputs will be presented through a lightweight dashboard designed solely for visual comparison and interpretation, ensuring a clear separation between pricing logic and result presentation.

Ethical Considerations

This study presents minimal ethical risk, as it uses publicly available, anonymised retail datasets and does not involve human participants or personal data. All data will be stored securely and accessed only by the researcher. The study does not influence real-world pricing decisions or interact with commercial environments, ensuring no impact on customers or businesses. Ethical considerations, therefore, focus on responsible data handling, transparent reporting of model behaviour and care in interpreting the practical relevance of the findings. As a computational study, it aligns with institutional ethical standards and is not expected to require additional ethical safeguards.

Timeline

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